

EDUCATION

University of North Carolina at Chapel Hill – Data Science and Statistics with Econometrics concentration 2023- 2027

PROFESSIONAL EXPERIENCE

PQLS – Tamil Nadu, India January 2026 – May 2026

Business Analytics Intern

- Analyzed 2K+ customer records across 35 variables with SQL and Excel to identify service usage patterns and improvement opportunities, building Tableau dashboards to support data-driven business decisions.

Adroitts – Reston, VA

June 2025 - August 2025

Data Science Intern

- Designed Power BI and Tableau dashboards to translate large marketing datasets into actionable insights, highlighting client KPIs for executive decision-making.

Carolina Housing – Chapel Hill, NC.

August 2024 - Present

Resident Advisor Mentor | May 2026–Present · Resident Advisor | August 2024–May 2026

- Supported 600+ residents, led crisis response & conflict mediation, & maintained compliance with university policy.

PROJECT AND RESEARCH EXPERIENCE

AI-Powered Word Game — IBM July Hackathon

September 2025

- Built a Flask-based application that retrieves Wikipedia content and uses the Gemini API to generate narrative summaries of historical periods, which is completed with three interactive wordplay clues.
- Developed a dataset-guided clue engine using 700 Parseword game examples, automated clue validation, and batched Gemini generation to improve clue consistency.

AI Summaries and User Engagement in Online Communities

May 2026 – Present

Undergraduate Researcher | Faculty Supervisor: Professor Fan Yao

- Investigate whether Reddit's AI-generated summaries substitute for user attention by reducing commenting and participation in the original human discussions.
- Develop the empirical design by operationalising treatment and engagement measures from public Reddit data and evaluating temporal variation, selection effects, measurement limitations, and causal-identification challenges

Tweet Sentiments During the COVID-19 Lockdown

January 2025 – April 2025

- Built a machine-learning pipeline integrating vaccine-related Twitter sentiment with vaccination and COVID-19 mortality data, preprocessed text and structured variables in Python.
- Compared Random Forest, SVM, KNN, and regression models, achieving an R^2 of 0.92 for daily mortality prediction and 37% multiclass sentiment accuracy using Random Forest.

Dollars in Orbit — Carolina Data Challenge 2025, 2nd Place Winner

September 2025

- Analyzed BEA space economy data from 2012–23 to quantify private and federal sector contributions to U.S. output.
- Developed Excel dashboards and KPI visualizations showing GDP contribution and funding impact.

PUBLICATIONS

Smart City Surveillance System

January 2022

International Journal for Innovative Research in Multidisciplinary Field

- Developed a conceptual framework for integrating public-service data across hospitals, police, traffic, schools, and municipal systems to improve response coordination and public safety.
- Explored how system-level data integration could strengthen service delivery and urban decision-making.

IoT-based Smart Waste Collector

August 2022

International Journal of Innovative Research in Science, Engineering and Technology

- Designed an IoT-enabled waste monitoring concept using sensors, web-based reporting, and automated alerts to improve waste collection efficiency.
- Focused on using embedded systems and data visibility to support cleaner and more responsive urban infrastructure.

SKILLS

Python, R, SQL, Excel, Tableau, Power BI, Flask, Gemini API, LLM Prompt Engineering, ML, Random Forest, SVM, NLP, Sentiment Analysis, Data Cleaning, Predictive Modelling